

Revenant Recovery Report

Deleted and superseded data remnants recovered from raw flash storage

Case	Demo Case 01
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Reference	MSc final project demo
Report generated	2026-09-23 21:17 UTC
Engine	Revenant 0.1.0
Case notes	LittleFS image built by the reference implementation, see littlefs_demo.img.truth.json.

Evidence

Source kind	image
Source	samples/littlefs_demo.img
Size	524,288 bytes
Acquired	2026-09-23T21:17:31.569170+00:00
Access	read only
SHA-256	4e522169f9b7cbca8a693a2f4cdc934c706223f9b5469a65bb93d8d861febaf1
MD5	1faacc87e85d7e8e83130191c0bb168b

The source was opened read only and hashed before analysis. The hashes above fix the state of the evidence at examination time, so the same image can be verified later.

Summary

The scan identified littlefs and examined 524,288 bytes in 0.01 seconds. It returned 6 artifacts in total, of which 3 belong to the live filesystem and 3 were recovered from obsolete, deleted, orphaned, or unallocated regions. The recovered items are the point of the exercise, since they represent data the device no longer presents through its filesystem. One signature find duplicated content that a filesystem parser had already recovered with a name and a state, so it was folded into that record.

State	Count	Meaning
deleted	2	removed from the filesystem, content recovered
obsolete	1	an earlier version later superseded
live	3	part of the active filesystem

Recovered items

Every row below is an item that is not part of the live filesystem. The offset points to the source record in the image and the hash fixes the recovered bytes.

Name	FS	State	Conf	Offset	Size	SHA-256 (first 16)
/log.txt	littlefs	obsolete	high	0x1194	60	18afd53e2486e6fd
/token.txt	littlefs	deleted	high	0x1224	46	5e777a66247fb451
/cam/snap_0001.jpg	littlefs	deleted	medium	0x33000	2,902	c09ac42be97bc1e7

Recovered content previews

A short printable preview of recovered items, with non printable bytes shown as dots. Deleted and obsolete items come first, since those carry the most weight.

/token.txt [deleted] littlefs 46 bytes

```
RVN-LDEMO-00003 SESSION_TOKEN=abcdef0123456789
```

removed by a delete tag, metadata block 1 revision 1

/cam/snap_0001.jpg [deleted] littlefs 2,902 bytes

```
......RVN-LDEMO-00005....JFIF.............C.............................................!........."$".$..
.....C.....................................................................@.`.."......................
.....................................}........!1A..Qa."q.2....#B...R..$3br........%&'()* ...
```

removed by a delete tag, ctz head block 51, metadata block 45 revision 0

/log.txt [obsolete] littlefs 60 bytes

```
RVN-LDEMO-00002 boot ok, user alice logged in from 10.0.0.23
```

superseded by a later commit, metadata block 1 revision 1

Method and limitations

Revenant reads the raw image and recovers remnants filesystem by filesystem. It never writes to the source. The notes below describe how far to trust each parser, which matters because some flash filesystems give up their history more readily than others.

littlefs. Replays every CRC checked commit, follows id shifts, and reads CTZ skip lists, then rebuilds unreferenced chains from their own back pointers. Scored against images made by the reference implementation.

Recovery figures should be read against a known ground truth. The controlled test images that ship with the tool have an exact record of what was deleted, so they give a fair measure of recall. Content pulled from real firmware only tells you the parser reads the format, not how much of a real deletion history it can bring back.

Chain of custody

The tool recorded each step of this examination with a timestamp. The trail holds sizes and hashes rather than recovered content, so it can be shared without exposing the data itself.

Time (UTC)	Action	Detail
2026-09-23T21:17:31	open_image	samples/littlefs_demo.img
2026-09-23T21:17:31	hash_image	sha256 4e522169f9b7cbca8a693a2f4cdc934c706223f9b5469a65bb93d8d861febaf1
2026-09-23T21:17:31	detect	littlefs
2026-09-23T21:17:31	carve_start	littlefs
2026-09-23T21:17:31	carve_done	littlefs: 6 artifacts
2026-09-23T21:17:31	carve_start	signatures
2026-09-23T21:17:31	carve_done	signatures: 1 artifacts
2026-09-23T21:17:31	merge	1 carved copies of already recovered content merged
2026-09-23T21:17:31	scan_complete	6 artifacts, 3 recovered